이론시험 문제
해설
주의: 채점표의 배점은 나눌 수 없음!
문제 1 (10.0점)
문제 1A (4.0점)

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. ,

$$v = const. \quad (1)$$

l .

2

$$T = m\frac{v^2}{l}. \quad (2)$$

$$dl = -Rd\alpha, \quad (3)$$

$$d\alpha = \omega dt, \quad (4)$$

$$\omega = \frac{v}{l}. \quad (5)$$

(3)-(5)

$$ldl = -Rvdt, \quad (6)$$

$$l^2 - l_0^2 = -2Rvt. \quad (7)$$

(1) (7) ,

$$t = \frac{l_0^2 - \left(\frac{mv^2}{T}\right)^2}{2Rv} = \frac{l_0^2T^2 - m^2v^4}{2RvT^2}. \quad (8)$$

평가 항목	배점
	1
$T = m\frac{v^2}{l}$	0.5
$ldl = -Rvdt$	1
$l^2 - l_0^2 = -2Rvt$	0.5
$t = \frac{l_0^2T^2 - m^2v^4}{2RvT^2}$	1
합계	**4.0**

문제 1B (3.0점)

가능한 풀이.

T T_x , α . ,

$$P = \alpha(T - T_x), \quad (1)$$

C dT

dt ,

$$CdT = -Pdt. \quad (2)$$

(1) (2) $T = T_0$

$$T(t) = T_x + (T_0 - T_x)e^{-\beta t}, \quad (3)$$

$\beta = \alpha / C$.